

Practice Exercises: Vectors and Applications

Mathematical Methods for Computing (CS-813)

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Instructions: Solve the following exercises clearly. Show the main computational steps. Where an application is given, briefly interpret the numerical result.

1. Basic Vector Operations

Exercise 1. Vector Addition – 2D

A robot moves first by

$$\mathbf{x} = (3, 2)$$

and then by

$$\mathbf{y} = (1, 4).$$

Find the total displacement

$$\mathbf{x} + \mathbf{y}.$$

Interpret the resulting vector geometrically.

Exercise 2. Vector Addition – 3D

Two feature vectors extracted from two different sources are

$$\mathbf{x} = (2, 1, 3), \quad \mathbf{y} = (1, 4, 2).$$

Find

$$\mathbf{x} + \mathbf{y}.$$

Interpret the answer as a combined feature vector.

Exercise 3. Vector Addition – 4D

Two parts of a user profile are represented by

$$\mathbf{x} = (2, 3, 1, 4), \quad \mathbf{y} = (1, 2, 3, 1).$$

Find their combined profile vector.

Exercise 4. Vector Subtraction – 2D

Two objects are located at

$$A = (1, 2), \quad B = (5, 4).$$

Find the displacement vector from A to B .

Exercise 5. Vector Subtraction – 3D

Given

$$\mathbf{x} = (5, 4, 2), \quad \mathbf{y} = (2, 1, 3),$$

find

$$\mathbf{x} - \mathbf{y}.$$

Which component has the largest absolute difference?

Exercise 6. Vector Subtraction – 4D

Two observations are

$$\mathbf{x} = (8, 4, 6, 2), \quad \mathbf{y} = (5, 3, 2, 1).$$

Find their difference vector.

Exercise 7. Scalar Multiplication – 2D

A point in a graphical object is represented by

$$\mathbf{x} = (2, 3).$$

The object is scaled by a factor of 2.

Find

$$2\mathbf{x}.$$

Does the line of action of the vector change?

Exercise 8. Scalar Multiplication – 3D

A feature vector is

$$\mathbf{x} = (4, 6, 2).$$

Find

$$\frac{1}{2}\mathbf{x}.$$

Describe what happens to its magnitude and direction.

Exercise 9. Scalar Multiplication – 4D

Let

$$\mathbf{x} = (2, 4, 6, 8).$$

Calculate

$$0.25\mathbf{x}.$$

Explain briefly how scalar multiplication can be used for feature weighting.

2. Dot Product and Norm**Exercise 10. Dot Product – 2D**

Let

$$\mathbf{x} = (2, 3), \quad \mathbf{y} = (4, 1).$$

Calculate

$$\mathbf{x} \cdot \mathbf{y}.$$

Exercise 11. Dot Product – 3D

Let

$$\mathbf{x} = (1, 2, 3), \quad \mathbf{y} = (2, 1, 2).$$

Calculate

$$\mathbf{x} \cdot \mathbf{y}.$$

Exercise 12. Dot Product – 4D

Let

$$\mathbf{x} = (1, 0, 2, 1), \quad \mathbf{y} = (2, 1, 1, 3).$$

Calculate

$$\mathbf{x} \cdot \mathbf{y}.$$

What does a positive dot product suggest about their general directional relationship?

Exercise 13. Euclidean Norm – 2D

Find the Euclidean norm of

$$\mathbf{x} = (3, 4).$$

Exercise 14. Euclidean Norm – 3D

Find

$$\|\mathbf{x}\|$$

for

$$\mathbf{x} = (2, 3, 6).$$

Exercise 15. Euclidean Norm – 4D

Calculate the length of

$$\mathbf{x} = (1, 2, 2, 4).$$

Exercise 16. Normalization

For

$$\mathbf{x} = (3, 4),$$

find the unit vector

$$\hat{\mathbf{x}} = \frac{\mathbf{x}}{\|\mathbf{x}\|}.$$

Verify that

$$\|\hat{\mathbf{x}}\| = 1.$$

3. Euclidean Distance**Exercise 17. Euclidean Distance – 2D**

A new customer has feature vector

$$Q = (4, 3).$$

Two existing customers are

$$A = (1, 3), \quad B = (5, 5).$$

Calculate

$$d(Q, A) \quad \text{and} \quad d(Q, B).$$

Which customer is more similar to Q ?

Exercise 18. Euclidean Distance – 3D

Let

$$P = (1, 2, 3), \quad Q = (3, 4, 6).$$

Calculate

$$d(P, Q).$$

Exercise 19. Euclidean Distance – 4D

Given

$$P = (1, 2, 1, 3), \quad Q = (3, 2, 4, 4),$$

calculate the Euclidean distance between them.

Exercise 20. Application: 1-Nearest-Neighbour Classification

A new data point is

$$Q = (3, 2).$$

The labelled training points are

$$A = (1, 2) \quad \text{Class 0,}$$

$$B = (2, 1) \quad \text{Class 0,}$$

$$C = (4, 4) \quad \text{Class 1,}$$

$$D = (5, 3) \quad \text{Class 1.}$$

Using Euclidean distance:

- a) Find $d(Q, A)$.
- b) Find $d(Q, B)$.
- c) Find $d(Q, C)$.
- d) Find $d(Q, D)$.
- e) Identify the nearest neighbour.
- f) Predict the class of Q .

4. Manhattan Distance

Exercise 21. Manhattan Distance – 2D

For

$$A = (1, 2), \quad B = (4, 6),$$

calculate

$$d_1(A, B) = |4 - 1| + |6 - 2|.$$

Exercise 22. Manhattan Distance – 3D

Let

$$P = (2, -1, 3), \quad Q = (5, 4, 1).$$

Find their Manhattan distance.

Exercise 23. Manhattan Distance – 4D

Find the Manhattan distance between

$$\mathbf{x} = (1, 3, 2, 5)$$

and

$$\mathbf{y} = (4, 1, 6, 3).$$

Exercise 24. Application: Manhattan Distance in k-NN

A new point is

$$Q = (5, 2).$$

The training data are

$$A = (1, 2) \quad \text{Class 0,}$$

$$B = (4, 1) \quad \text{Class 0,}$$

$$C = (3, 5) \quad \text{Class 1,}$$

$$D = (6, 4) \quad \text{Class 1.}$$

Using Manhattan distance:

- Calculate the distance from Q to every training point.
- Identify the nearest neighbour.
- Predict the class of Q .

5. Angle and Cosine Similarity

Recall that

$$\cos \theta = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}.$$

Exercise 25. Angle Between Vectors – 2D

Find the angle between

$$\mathbf{x} = (2, 1), \quad \mathbf{y} = (1, 2).$$

Exercise 26. Angle Between Vectors – 3D

Find the angle between

$$\mathbf{x} = (1, 1, 0), \quad \mathbf{y} = (1, 0, 1).$$

Exercise 27. Angle Between Vectors – 4D

Find the cosine of the angle between

$$\mathbf{x} = (1, 1, 0, 0)$$

and

$$\mathbf{y} = (1, 0, 1, 0).$$

Hence determine the angle.

Exercise 28. Application: Document Similarity

Suppose the vocabulary is

$$\{\text{data, science, machine}\}.$$

Two documents are represented by

$$D_1 = (2, 1, 0), \quad D_2 = (1, 0, 1).$$

Calculate:

a) $D_1 \cdot D_2$,

b) $\|D_1\|$,

c) $\|D_2\|$,

d) cosine similarity

$$\cos(D_1, D_2),$$

e) the angle between D_1 and D_2 .

Interpret whether the documents have low, moderate, or high directional similarity.

6. Orthogonal and Orthonormal Vectors

Exercise 29. Orthogonal Vectors – 2D

Determine whether

$$\mathbf{x} = (1, 2), \quad \mathbf{y} = (2, -1)$$

are orthogonal.

Exercise 30. Orthogonal Vectors – 3D

Determine whether

$$\mathbf{x} = (1, 2, -1), \quad \mathbf{y} = (2, -1, 0)$$

are orthogonal.

Exercise 31. Orthogonal Vectors – 4D

Determine whether

$$\mathbf{x} = (1, 1, 1, 1), \quad \mathbf{y} = (1, -1, 1, -1)$$

are orthogonal.

Exercise 32. Orthonormal Vectors – 2D

Verify that

$$\mathbf{e}_1 = (1, 0), \quad \mathbf{e}_2 = (0, 1)$$

form an orthonormal set.

Exercise 33. Orthonormal Vectors – 3D

Verify that

$$\mathbf{e}_1 = (1, 0, 0), \quad \mathbf{e}_2 = (0, 1, 0), \quad \mathbf{e}_3 = (0, 0, 1)$$

are mutually orthonormal.

Exercise 34. Orthonormal Vectors – 4D

Consider

$$\begin{aligned} \mathbf{e}_1 &= (1, 0, 0, 0), & \mathbf{e}_2 &= (0, 1, 0, 0), \\ \mathbf{e}_3 &= (0, 0, 1, 0), & \mathbf{e}_4 &= (0, 0, 0, 1). \end{aligned}$$

Verify that they form an orthonormal basis of

$$\mathbb{R}^4.$$

7. Application Problems

Exercise 35. Recommendation System

A user's preferences for three movie genres are

$$U = (5, 3, 1).$$

Two movies are represented by

$$M_1 = (4, 3, 1), \quad M_2 = (1, 2, 5).$$

a) Calculate

$$d(U, M_1) \quad \text{and} \quad d(U, M_2).$$

b) Which movie is closer to the user's preferences?

c) Which movie should be recommended?

d) Compare the movies again using cosine similarity.

Exercise 36. Image Classification

A new image is represented by the feature vector

$$Q = (3, 4).$$

The training images are

$$A = (1, 2) \quad \text{Cat},$$

$$B = (4, 5) \quad \text{Dog},$$

$$C = (6, 2) \quad \text{Dog}.$$

Using Euclidean distance:

a) Calculate the distance from Q to each image.

b) Identify the nearest training image.

c) Predict the class of Q using 1-NN.

Exercise 37. Text Classification

A text message is represented by

$$Q = (2, 1, 3).$$

Two representative feature vectors are

$$S = (3, 1, 4) \quad \text{Spam},$$

$$N = (0, 3, 1) \quad \text{Not Spam}.$$

Using Manhattan distance:

a) Calculate

$$d_1(Q, S).$$

b) Calculate

$$d_1(Q, N).$$

c) Predict the class of Q .

Exercise 38. Compare Euclidean and Manhattan Distance

Let

$$Q = (3, 3), \quad A = (1, 2), \quad B = (4, 5).$$

a) Calculate

$$d_2(Q, A) \quad \text{and} \quad d_2(Q, B).$$

b) Calculate

$$d_1(Q, A) \quad \text{and} \quad d_1(Q, B).$$

c) Determine the nearest point under each metric.

d) Do both distance measures produce the same nearest neighbour?

8. Integrated Problem

Exercise 39. Integrated Vector Problem in $\mathbb{R}^4$

Consider

$$\mathbf{x} = (1, 2, 0, 2), \quad \mathbf{y} = (2, 1, 2, 0).$$

Calculate:

a)

$$\mathbf{x} + \mathbf{y}.$$

b)

$$\mathbf{x} - \mathbf{y}.$$

c)

$$\mathbf{x} \cdot \mathbf{y}.$$

d)

$$\|\mathbf{x}\| \quad \text{and} \quad \|\mathbf{y}\|.$$

e) Euclidean distance

$$\|\mathbf{x} - \mathbf{y}\|.$$

f) Manhattan distance

$$d_1(\mathbf{x}, \mathbf{y}).$$

g) Cosine similarity

$$\cos(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}.$$

h) The angle between

$\mathbf{x}$ and $\mathbf{y}$.

i) Determine whether the vectors are orthogonal.

9. Conceptual Application

Exercise 40. Choosing a Suitable Distance or Similarity Measure

For each situation, choose an appropriate mathematical measure and briefly justify your answer.

a) Two points represent physical positions in a plane.

Choose between:

Euclidean distance and cosine similarity.

b) Documents are represented by high-dimensional word-frequency vectors, and direction matters more than magnitude.

c) A robot moves on a rectangular grid and is allowed to move only horizontally or vertically.

Choose between:

Euclidean distance and Manhattan distance.

d) Two embedding vectors represent the semantic meaning of two sentences.

Which measure would you prefer for comparing their directions?

Mathematics provides the language for representing, comparing and understanding data.